Mandapalli Rashmika

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	6.30
Lakshya International Schools	Class XII (CBSE)	2023	88.00%
Aditya High School	Class X (SSC)	2021	9.80%

WORK EXPERIENCE

Summer Intern : Prof Praveen Tammana

- Developed a Hackathon Evaluation & Verification Engine combining static code analysis, complexity/testing checks, and problem statement coverage mapping to produce evidence backed technical evaluations.
- Built a sandboxed dynamic validation pipeline to safely execute untrusted submissions, exercise API endpoints, verify data flow, and detect hardcoded or non-functional implementations using isolated, resource-limited containers.
- Integrated Gemini based evaluation to generate structured technical assessments with supporting evidence, identify critical red flags, and formulate targeted probing questions for jury review.
- Built an AI Research Navigator over a curated 50-document corpus spanning LLMs, RAG, agents, alignment, and evaluation, using normalized metadata and topic tags to organize technical research resources

PROJECTS

Notification Framework

- Built a full-stack multi-channel notification platform Implemented secure user authentication
- Added rate-limiting and client management features
- Developed webhook and media-handling capabilities

Intent-Based Networking (IBN)

- Designed and implemented an Intent-Based Networking (IBN) system using P4, SDN, and Mininet.
- Integrated a RASA-based NLU pipeline for intent classification and entity extraction.
- Enabled dynamic policy enforcement by integrating an SDN controller that computes forwarding paths using Dijkstra's algorithm and installs rules on programmable switches via P4Runtime.

Rental Management System

- Collaborated on a full stack platform for managing properties, tenants, leases, payments, and maintenance efficiently.
- Equipped tenants to view listings, request maintenance, pay rent online and admins with tools to oversee
- Developed a rental management database using PostgreSQL for properties, tenants, leases, payments, and maintenance.
- Implemented workflows for tenant registration, lease management, payment tracking, and maintenance requests.

ProbScript DSL Compiler

- Designed a DSL for probabilistic programming with native support for distributions and random variables.
- Built a complete compiler frontend using Flex, Bison, semantic analysis, type checking, and symbol resolution.
- Developed a rich type system supporting vectors, matrices, sets, maps, probabilistic types, type inference, and coercion

RISC-V Processor Simulator

- Engineered a RISC-V simulator supporting assembly, instruction execution, memory management, and debugging.
- Implemented a configurable cache system with write-back/write-through policies and performance metrics.
- Developed instruction pipelining concepts to improve execution efficiency and reduce clock cycles.

SKILLS

- **Languages** : C, C++, JavaScript, Assembly, Python
- **Web Technologies** : HTML, CSS, ReactJS, FastAPI
- **Databases** : MySQL, PostgresSQL
- **Other Tools** : LaTeX, Shell Scripting, Linux, CMake, Flex, Bison
- **DevOps** : Docker, Git, GitHub
- **Libraries** : NumPy, Pandas, Matplotlib

RELEVANT COURSES

Theory and Algorithms : Data Structures , Algorithms , Compilers , Database Management Systems , Theory of Computation , Approximation Algorithms

Mathematics and Statistics : Probability and Random Variables, Statistics, Linear Algebra, Calculus, Discrete Mathematics

Systems & Networks : Operating Systems, Computer Architecture, Computer Networks, Database Management Systems, Software EngineeringSystems & Networks

Specialized : Foundations of Machine Learning , Deep learning , Fraud Analytics

SCHOLASTIC ACHIEVEMENTS

Qualified for the NTSE (National Talent Search Examination) and have been receiving the associated scholarship since the first year of my intermediate education

EXTRACURRICULAR

- Member of National Social Service , IIT Hyderabad(2023-27)
- Actively participate in voluteering events